

## Scalability & Future Plan

|                      |                                 |
|----------------------|---------------------------------|
| <b>Date</b>          | 04 July 2026                    |
| <b>Team ID</b>       | XXXXXX                          |
| <b>Project Name</b>  | Internal Job Mobility Assistant |
| <b>Maximum Marks</b> | 1 Mark                          |

### Scalability

| Aspect                 | Details                                                                                                                                                                                                                                                                         |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Current Architecture   | The application follows a modular client-server architecture with a React frontend deployed on Vercel and a FastAPI backend deployed on Render. AI services are integrated using Google Gemini API, while SQLite with SQLAlchemy ORM manages application data.                  |
| Scalability Approach   | The modular architecture allows independent scaling of the frontend and backend. The backend can be migrated from SQLite to PostgreSQL or MySQL for larger datasets, and API services can be containerized and deployed on cloud infrastructure to support higher user traffic. |
| Expected Growth / Load | The current implementation is suitable for small and medium-sized organizations. With cloud database migration, caching, and load balancing, the system can support thousands of employees and concurrent job applications efficiently.                                         |

### Future Plan

| S.No | Planned Feature / Enhancement                                                             | Target Timeline           | Priority |
|------|-------------------------------------------------------------------------------------------|---------------------------|----------|
| 1    | AI-powered interview preparation with personalized mock interview questions and feedback  | Next Release              | High     |
| 2    | Advanced HR analytics dashboard with predictive hiring insights and workforce trends      | Next 3–6 Months           | High     |
| 3    | Integration with enterprise HRMS, LDAP, and Single Sign-On (SSO) systems                  | Future Enterprise Release | Medium   |
| 4    | Mobile application with real-time notifications, application tracking, and career updates | Future Release            | Medium   |